

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

Semester-VIII of B. Pharm. Examination

University Backlog Theory Examination November - 2016

PH411 Pharmaceutical Analysis-IV

Date: 09/11/2016, Wednesday Time: 01:30 p.m. To 04:30 p.m. Maximum Marks: 80

Instructions:

1. Figures to right indicate marks.
2. Section wise separate answer book is to be used.
3. Draw neat sketches wherever necessary.

SECTION- I

- Q 1** Attempt any TWO of the following:
- A Write a Short Note On:**
- I. Give the difference between AAS and AES 05
 - II. Discuss the interference in Atomic Absorption Spectroscopy 05
- B Write a Short Note On:**
- I. Give statement of Bragg's Law and its derivation 05
 - II. Enlist X – ray diffraction techniques and discuss in detail Debye Scherrer Method. 05
- C** Draw the schematic diagram of instrument for Super critical fluid Chromatography. 10
Discuss advantages, limitations and applications of Super Critical Fluid Chromatography
- Q 2** Attempt any TWO of the following:
- A Write a Short Note On:**
- I. Define Super Critical Fluid. Discuss properties and modifiers used for Super Critical Fluid. 05
 - II. Enumerate detectors used in X-ray Crystallography and discuss any one in detail 05
- B Write a Short Note On:**
- I. Write principle of Atomic absorption Spectroscopy. Give its applications. 05
 - II. Discuss sources for generation of X-rays 05
- C Write a Short Note On:**
- I. Describe the applications and recent advancement of X- ray Crystallography 05
 - II. Write a short note on Flame photometer. 05

SECTION- II

- Q 3** Attempt any TWO of the following:
- A**
- I. Explain the Principle of NMR Spectroscopy. 04
 - II. Spin-Spin Coupling 03
 - III. Shielding and Deshielding effect 03

B	I.	Write a Short Note On:	05
		Detector Used in Mass Spectroscopy	05
C	II.	Enumerate Applications of Mass Spectroscopy.	05
		Write a detailed note on Fast Atom Bombardment Mass Spectroscopy (FAB-MASS).	10
Q 4		Attempt any <u>TWO</u> of the following:	
		What is Chemical Shift? Write brief note on factors influencing chemical shift.	10
B	I.	Write a Short Note On:	
		Enumerate Applications of NMR Spectroscopy.	05
	II.	Isotopic Abundance	05
		Write a Short Note On:	05
C	I.	Coupling Constant	05
		Metastable Ions or Peaks	05